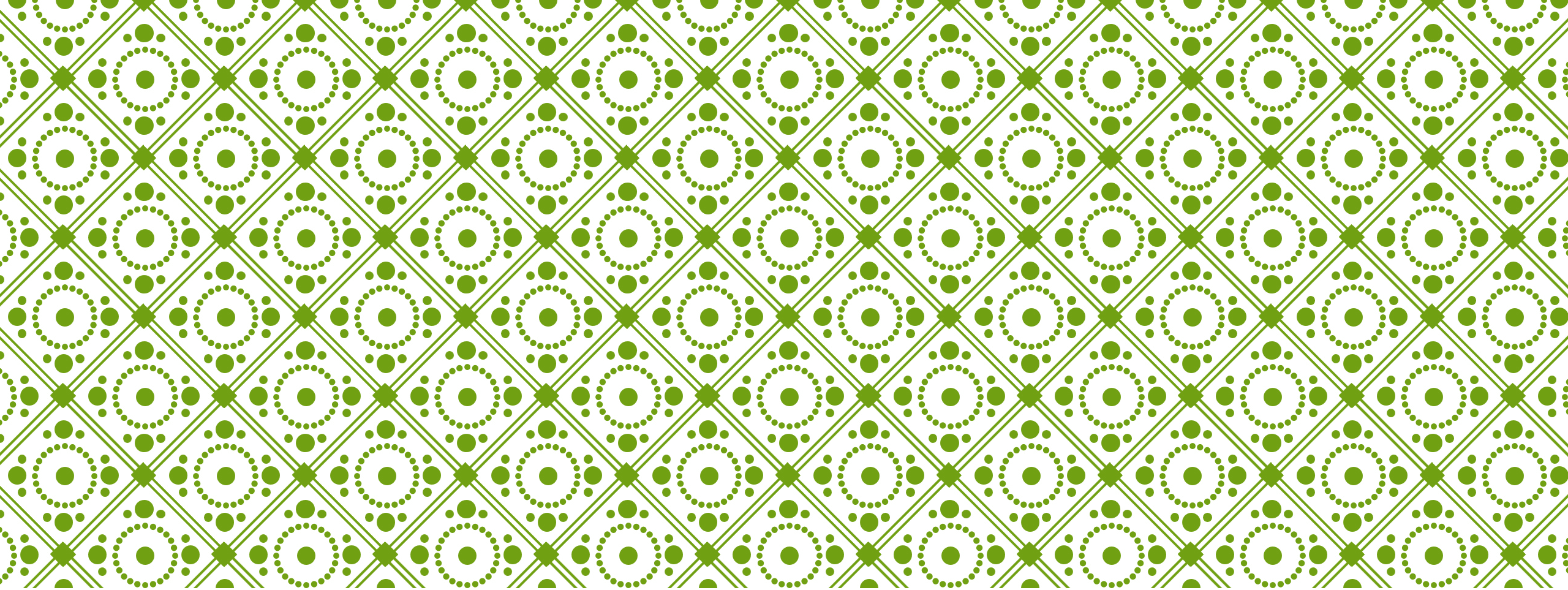


CONTINUOUS PROBABILITY DISTRIBUTIONS

PK

For borrowers with good credit scores, the mean debt for revolving and installment accounts is $15,015 (*BusinessWeek*, March 20, 2006). Assume the standard deviation is $3540 and that debt amounts are normally distributed.

a. What is the probability that the debt for a borrower with good credit is more than $18,000?

b. What is the probability that the debt for a borrower with good credit is less than $10,000?

c. What is the probability that the debt for a borrower with good credit is between $12,000 and $18,000?

d. What is the probability that the debt for a borrower with good credit is no more than $14,000?

The average return for large-cap domestic stock funds over the three years 2009–2011 was 14.4% (*AAII Journal*, February, 2012). Assume the three-year returns were normally distributed across funds with a standard deviation of 4.4%.

a. What is the probability an individual large-cap domestic stock fund had a three-year return of at least 20%?

b. What is the probability an individual large-cap domestic stock fund had a three-year return of 10% or less?

c. How big does the return have to be to put a domestic stock fund in the top 10% for the three-year period?

In an article about the cost of health care, *Money* magazine reported that a visit to a hospital emergency room for something as simple as a sore throat has a mean cost of $328 (*Money*, January 2009). Assume that the cost for this type of hospital emergency room visit is normally distributed with a standard deviation of $92. Answer the following questions about the cost of a hospital emergency room visit for this medical service.

a. What is the probability that the cost will be more than $500?

b. What is the probability that the cost will be less than $250?

c. What is the probability that the cost will be between $300 and $400?

d. If the cost to a patient is in the lower 8% of charges for this medical service, what was the cost of this patient's emergency room visit?

Television viewing reached a new high when the Nielsen Company reported a mean daily viewing time of 8.35 hours per household (*USA Today*, November 11, 2009). Use a normal probability distribution with a standard deviation of 2.5 hours to answer the following questions about daily television viewing per household.

a. What is the probability that a household views television between 5 and 10 hours a day?

b. How many hours of television viewing must a household have in order to be in the top 3% of all television viewing households?

c. What is the probability that a household views television more than 3 hours a day?

According to the Sleep Foundation, the average night's sleep is 6.8 hours (*Fortune*, March 20, 2006). Assume the standard deviation is .6 hours and that the probability distribution is normal.

a. What is the probability that a randomly selected person sleeps more than 8 hours?

b. What is the probability that a randomly selected person sleeps 6 hours or less?

c. Doctors suggest getting between 7 and 9

The average price for a gallon of gasoline in the United States is $3.73 and in Russia it is $3.40 (*Bloomberg Businessweek*, March 5–March 11, 2012). Assume these averages are the population means in the two countries and that the probability distributions are normally distributed with a standard deviation of $.25 in the United States and a standard deviation of $.20 in Russia.

a. What is the probability that a randomly selected gas station in the United States charges less than $3.50 per gallon?

b. What percentage of the gas stations in Russia charge less than $3.50 per gallon?

c. What is the probability that a randomly selected gas station in Russia charged more than the mean price in the United States?

The time needed to complete a final examination in a particular college course is normally distributed with a mean of 80 minutes and a standard deviation of 10 minutes. Answer the following questions.

a. What is the probability of completing the exam in one hour or less?

b. What is the probability that a student will complete the exam in more than 60 minutes but less than 75 minutes?

c. Assume that the class has 60 students and that the examination period is 90 minutes in length. How many students do you expect will be unable to complete the exam in the allotted time?

Trading volume on the New York Stock Exchange is heaviest during the first half hour (early morning) and last half hour (late afternoon) of the trading day. The early morning trading volumes (millions of shares) for 13 days in January and February are shown here (*Barron's*, January 23, 2006; February 13, 2006; and February 27, 2006).

214	163	265	194	180
202	198	212	201	
174	171	211	211	

The probability distribution of trading volume is approximately normal.

a. Compute the mean and standard deviation to use as estimates of the population mean and standard deviation.
b. What is the probability that, on a randomly selected day, the early morning trading volume will be less than 180 million shares?
c. What is the probability that, on a randomly selected day, the early morning trading volume will exceed 230 million shares?
d. How many shares would have to be traded for the early morning trading volume on a particular day to be among the busiest 5% of days?